

Recent-onset chest pain of suspected cardiac origin: assessment and diagnosis

Clinical guideline

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Your responsibility

The recommendations in this guideline represent the view of NICE, arrived at after careful consideration of the evidence available. When exercising their judgement, professionals and practitioners are expected to take this guideline fully into account, alongside the individual needs, preferences and values of their patients or the people using their service. It is not mandatory to apply the recommendations, and the guideline does not override the responsibility to make decisions appropriate to the circumstances of the individual, in consultation with them and their families and carers or guardian.

All problems (adverse events) related to a medicine or medical device used for treatment or in a procedure should be reported to the Medicines and Healthcare products Regulatory Agency using the [Yellow Card Scheme](#).

Local commissioners and providers of healthcare have a responsibility to enable the guideline to be applied when individual professionals and people using services wish to use it. They should do so in the context of local and national priorities for funding and developing services, and in light of their duties to have due regard to the need to eliminate unlawful discrimination, to advance equality of opportunity and to reduce health inequalities. Nothing in this guideline should be interpreted in a way that would be inconsistent with complying with those duties.

Commissioners and providers have a responsibility to promote an environmentally sustainable health and care system and should [assess and reduce the environmental impact of implementing NICE recommendations](#) wherever possible.

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This guideline partially replaces TA73.

This guideline is the basis of QS21 and QS68.

This guideline should be read in conjunction with NG185.

Overview

This guideline covers assessing and diagnosing recent chest pain in people aged 18 and over and managing symptoms while a diagnosis is being made. It aims to improve outcomes by providing advice on tests (ECG, high-sensitivity troponin tests, multislice CT angiography, functional testing) that support healthcare professionals to make a speedy and accurate diagnosis.

NICE has also produced a [guideline on managing acute coronary syndromes](#).

Who is it for?

- Healthcare professionals
- Adults with chest pain of recent onset, their families and carers

Using this guideline

People have the right to be involved in discussions and make informed decisions about their care, as described in [NICE's information on making decisions about your care](#).

Healthcare professionals should follow our general guidelines for people delivering care:

- [Decision-making and mental capacity](#)
- [Medicines adherence](#)
- [Medicines optimisation](#)
- [Multimorbidity](#)
- [Patient experience in adult NHS services](#)
- [Shared decision making](#)

[Making decisions using NICE guidelines](#) explains how we use words to show the strength (or certainty) of our recommendations, and has information about prescribing medicines (including off-label use), professional guidelines, standards and laws (including on consent and mental capacity), and safeguarding.

Recommendations

1.1 Providing information for people with chest pain

- 1.1.1.1 Discuss any concerns people (and where appropriate their family or carer/ advocate) may have, including anxiety when the cause of the chest pain is unknown. Correct any misinformation. **[2010]**
- 1.1.1.2 Offer people a clear explanation of the possible causes of their symptoms and the uncertainties. **[2010]**
- 1.1.1.3 Clearly explain the options to people at every stage of investigation. Make joint decisions with them and take account of their preferences:
- Encourage people to ask questions.
 - Provide repeated opportunities for discussion.
 - Explain test results and the need for any further investigations. **[2010]**
- 1.1.1.4 Provide information about any proposed investigations using everyday, jargon-free language. Include:
- their purpose, benefits and any limitations of their diagnostic accuracy
 - duration
 - level of discomfort and invasiveness
 - risk of adverse events. **[2010]**
- 1.1.1.5 Offer information about the risks of diagnostic testing, including any radiation exposure. **[2010]**
- 1.1.1.6 Address any physical or learning difficulties, sight or hearing problems and difficulties with speaking or reading English, which may affect people's understanding of the information offered. **[2010]**

- 1.1.1.7 Offer information after diagnosis as recommended in the relevant disease management guidelines:
- [NICE guideline on acute coronary syndromes](#)
 - [NICE guideline on stable angina](#)
 - [NICE guideline on generalised anxiety disorder and panic disorder in adults](#)
 - [NICE guideline on gastro-oesophageal reflux disease and dyspepsia in adults](#). **[2010]**
- 1.1.1.8 Explain if the chest pain is non-cardiac and refer people for further investigation if appropriate. **[2010]**
- 1.1.1.9 Provide individual advice to people about seeking medical help if they have further chest pain. **[2010]**

1.2 People presenting with acute chest pain

This section of the guideline covers the assessment and diagnosis of people with recent acute chest pain or discomfort, suspected to be caused by an acute coronary syndrome (ACS). The term ACS covers a range of conditions including unstable angina, ST-segment-elevation myocardial infarction (STEMI) and non-ST-segment-elevation myocardial infarction (NSTEMI).

The guideline addresses assessment and diagnosis irrespective of setting, because people present in different ways. For early management of these conditions, follow the [NICE guideline on acute coronary syndromes](#).

1.2.1 Initial assessment and referral to hospital

- 1.2.1.1 Check immediately whether people currently have chest pain. If they are pain free, check when their last episode of pain was, particularly if they have had pain in the last 12 hours. **[2010]**
- 1.2.1.2 Determine whether the chest pain may be cardiac and therefore whether this

guideline is relevant, by considering:

- the history of the chest pain
- the presence of cardiovascular risk factors
- history of ischaemic heart disease and any previous treatment
- previous investigations for chest pain. **[2010]**

1.2.1.3 Initially assess people for any of the following symptoms, which may indicate an ACS:

- pain in the chest and/or other areas (for example, the arms, back or jaw) lasting longer than 15 minutes
- chest pain associated with nausea and vomiting, marked sweating, breathlessness, or particularly a combination of these
- chest pain associated with haemodynamic instability
- new onset chest pain, or abrupt deterioration in previously stable angina, with recurrent chest pain occurring frequently and with little or no exertion, and with episodes often lasting longer than 15 minutes. **[2010]**

1.2.1.4 Do not use people's response to glyceryl trinitrate (GTN) to make a diagnosis. **[2010]**

1.2.1.5 Do not assess symptoms of an ACS differently in men and women. Not all people with an ACS present with central chest pain as the predominant feature. **[2010]**

1.2.1.6 Do not assess symptoms of an ACS differently in ethnic groups. There are no major differences in symptoms of an ACS among different ethnic groups. **[2010]**

1.2.1.7 Refer people to hospital as an emergency if an ACS is suspected (see recommendation 1.2.1.3) and:

- they currently have chest pain **or**
- they are currently pain free, but had chest pain in the last 12 hours, and a resting 12-lead ECG is abnormal or not available. **[2010]**

1.2.1.8 If an ACS is suspected (see recommendation 1.2.1.3) and there are no reasons for emergency referral, refer people for urgent same-day assessment if:

- they had chest pain in the last 12 hours, but are now pain free with a normal resting 12-lead ECG **or**
- the last episode of pain was 12 to 72 hours ago. **[2010]**

1.2.1.9 Refer people for assessment in hospital if an ACS is suspected (see recommendation 1.2.1.3) and:

- the pain has resolved **and**
- there are signs of complications such as pulmonary oedema.

Use clinical judgement to decide whether referral should be as an emergency or urgent same-day assessment. **[2010]**

1.2.1.10 If a recent ACS is suspected in people whose last episode of chest pain was more than 72 hours ago and who have no complications such as pulmonary oedema:

- carry out a detailed clinical assessment (see recommendations 1.2.4.2 and 1.2.4.3)
- confirm the diagnosis by resting 12-lead ECG and blood troponin level
- take into account the length of time since the suspected ACS when interpreting the troponin level.

Use clinical judgement to decide whether referral is necessary and how urgent this should be. **[2010]**

1.2.1.11 Refer people to hospital as an emergency if they have a recent (confirmed or suspected) ACS and develop further chest pain. **[2010]**

1.2.1.12 When an ACS is suspected, start management immediately in the order appropriate to the circumstances (see the section on immediate management of a suspected acute coronary syndrome) and take a resting 12-lead ECG (see the section on resting 12-lead ECG). Take the ECG as soon as possible, but do not delay transfer to hospital. **[2010]**

- 1.2.1.13 If an ACS is not suspected, consider other causes of the chest pain, some of which may be life-threatening (see recommendations 1.2.6.5, 1.2.6.7 and 1.2.6.8). **[2010]**

1.2.2 Resting 12-lead ECG

- 1.2.2.1 Take a resting 12-lead ECG as soon as possible. When people are referred, send the results to hospital before they arrive if possible. Recording and sending the ECG should not delay transfer to hospital. **[2010]**
- 1.2.2.2 Follow local protocols for people with a resting 12-lead ECG showing regional ST-segment elevation or presumed new left bundle branch block (LBBB) consistent with an acute STEMI until a firm diagnosis is made. Continue to monitor (see recommendation 1.2.3.4). **[2010]**
- 1.2.2.3 Follow the [NICE guideline on acute coronary syndromes](#) for people with a resting 12-lead ECG showing regional ST-segment depression or deep T wave inversion suggestive of a NSTEMI or unstable angina until a firm diagnosis is made. Continue to monitor (see recommendation 1.2.3.4). **[2010]**
- 1.2.2.4 Even in the absence of ST-segment changes, have an increased suspicion of an ACS if there are other changes in the resting 12-lead ECG, specifically Q waves and T wave changes. Consider following the [NICE guideline on acute coronary syndromes](#) if these conditions are likely. Continue to monitor (see recommendation 1.2.3.4). **[2010]**
- 1.2.2.5 Do not exclude an ACS when people have a normal resting 12-lead ECG. **[2010]**
- 1.2.2.6 If a diagnosis of ACS is in doubt, consider:
- taking serial resting 12-lead ECGs
 - reviewing previous resting 12-lead ECGs
 - recording additional ECG leads.

Use clinical judgement to decide how often this should be done. Note that

the results may not be conclusive. **[2010]**

- 1.2.2.7 Obtain a review of resting 12-lead ECGs by a healthcare professional qualified to interpret them as well as taking into account automated interpretation. **[2010]**
- 1.2.2.8 If clinical assessment (as described in recommendation 1.2.1.10) and a resting 12-lead ECG make a diagnosis of ACS less likely, consider other acute conditions. First consider those that are life-threatening such as pulmonary embolism, aortic dissection or pneumonia. Continue to monitor (see recommendation 1.2.3.4). **[2010]**

1.2.3 Immediate management of a suspected acute coronary syndrome

Management of ACS should start as soon as it is suspected, but should not delay transfer to hospital. The recommendations in this section should be carried out in the order appropriate to the circumstances.

- 1.2.3.1 Offer pain relief as soon as possible. This may be achieved with GTN (sublingual or buccal), but offer intravenous opioids such as morphine, particularly if an acute myocardial infarction (MI) is suspected. **[2010]**
- 1.2.3.2 Offer people a single loading dose of 300 mg aspirin as soon as possible unless there is clear evidence that they are allergic to it.

If aspirin is given before arrival at hospital, send a written record that it has been given with the person.

Only offer other antiplatelet agents in hospital. Follow the [NICE guideline on acute coronary syndromes](#). **[2010]**

- 1.2.3.3 Do not routinely administer oxygen, but monitor oxygen saturation using pulse oximetry as soon as possible, ideally before hospital admission. Only offer supplemental oxygen to:
- people with oxygen saturation (SpO₂) of less than 94% who are not at risk of hypercapnic respiratory failure, aiming for SpO₂ of 94% to 98%

- people with chronic obstructive pulmonary disease who are at risk of hypercapnic respiratory failure, to achieve a target SpO₂ of 88% to 92% until blood gas analysis is available. **[2010]**

Be aware that some pulse oximeters can underestimate or overestimate oxygen saturation levels, especially if the saturation level is borderline. Overestimation has been reported in people with dark skin. See also the [NHS England Patient Safety Alert on the risk of harm from inappropriate placement of pulse oximeter probes](#).

1.2.3.4 Monitor people with acute chest pain, using clinical judgement to decide how often this should be done, until a firm diagnosis is made. This should include:

- exacerbations of pain and/or other symptoms
- pulse and blood pressure
- heart rhythm
- oxygen saturation by pulse oximetry
- repeated resting 12-lead ECGs **and**
- checking pain relief is effective. **[2010]**

1.2.3.5 Manage other therapeutic interventions using the [NICE guideline on acute coronary syndromes](#). **[2010]**

1.2.4 Assessment in hospital for people with a suspected acute coronary syndrome

1.2.4.1 Take a resting 12-lead ECG and a blood sample for high-sensitivity troponin I or T measurement (see the [section on use of biochemical markers for diagnosis of an acute coronary syndrome](#)) on arrival in hospital. **[2010, amended 2016]**

1.2.4.2 Carry out a physical examination to determine:

- haemodynamic status

- signs of complications, for example, pulmonary oedema, cardiogenic shock **and**
- signs of non-coronary causes of acute chest pain, such as aortic dissection. **[2010]**

1.2.4.3 Take a detailed clinical history unless a STEMI is confirmed from the resting 12-lead ECG (that is, regional ST-segment elevation or presumed new LBBB). Record:

- the characteristics of the pain
- other associated symptoms
- any history of cardiovascular disease
- any cardiovascular risk factors **and**
- details of previous investigations or treatments for similar symptoms of chest pain. **[2010]**

1.2.5 Use of biochemical markers for diagnosis of an acute coronary syndrome

1.2.5.1 Do not use high-sensitivity troponin tests for people in whom ACS is not suspected. **[2016]**

1.2.5.2 For people at high or moderate risk of MI (as indicated by a validated tool), perform high-sensitivity troponin tests as recommended in the [NICE diagnostics guidance on myocardial infarction](#). **[2016]**

1.2.5.3 For people at low risk of MI (as indicated by a validated tool):

- perform a second high-sensitivity troponin test as recommended in the [NICE diagnostics guidance on myocardial infarction](#) if the first troponin test at presentation is positive.
- consider performing a single high-sensitivity troponin test only at presentation to rule out NSTEMI if the first troponin test is below the lower

limit of detection (negative). **[2016]**

- 1.2.5.4 Ensure that patients understand that a detectable troponin on the first high-sensitivity test does not necessarily indicate that they have had an MI. **[2016]**
- 1.2.5.5 Do not use biochemical markers such as natriuretic peptides and high-sensitivity C-reactive protein to diagnose an ACS. **[2010]**
- 1.2.5.6 Do not use biochemical markers of myocardial ischaemia (such as ischaemia-modified albumin) as opposed to markers of necrosis when assessing people with acute chest pain. **[2010]**
- 1.2.5.7 When interpreting high-sensitivity troponin measurements, take into account:
- the clinical presentation
 - the time from onset of symptoms
 - the resting 12-lead ECG findings
 - the pre-test probability of NSTEMI
 - the length of time since the suspected ACS
 - the probability of chronically elevated troponin levels in some people
 - that 99th percentile thresholds for troponin I and T may differ between sexes. **[2010, amended 2016]**

1.2.6 Making a diagnosis

- 1.2.6.1 When diagnosing MI, use the universal definition of myocardial infarction. This is the detection of rise and/or fall of cardiac biomarkers values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile of the upper reference limit and at least one of the following:
- symptoms of ischaemia
 - new or presumed new significant ST-segment-T wave (ST-T) changes or new

LBBB

- development of pathological Q waves in the ECG
- imaging evidence of new loss of viable myocardium or new regional wall motion abnormality
- identification of an intracoronary thrombus by angiography. **[2010, amended 2016]**

1.2.6.2 When a raised troponin level is detected in people with a suspected ACS, reassess to exclude other causes for raised troponin (for example, myocarditis, aortic dissection or pulmonary embolism) before confirming the diagnosis of ACS. **[2010]**

1.2.6.3 When a raised troponin level is detected in people with a suspected ACS, follow the [NICE guideline on acute coronary syndromes](#) until a firm diagnosis is made. Continue to monitor (see recommendation 1.2.3.4). **[2010]**

1.2.6.4 When a diagnosis of ACS is confirmed, follow the [NICE guideline on acute coronary syndromes](#). **[2010]**

1.2.6.5 Reassess people with chest pain without raised troponin levels and no acute resting 12-lead ECG changes to determine whether their chest pain is likely to be cardiac.

If myocardial ischaemia is suspected, follow the recommendations on stable chest pain in this guideline (see the [section on people presenting with stable chest pain](#)). Use clinical judgement to decide on the timing of any further diagnostic investigations. **[2010, amended 2016]**

1.2.6.6 Do not routinely offer non-invasive imaging or exercise ECG in the initial assessment of acute cardiac chest pain. **[new 2016]**

1.2.6.7 Only consider early chest computed tomography (CT) to rule out other diagnoses such as pulmonary embolism or aortic dissection, not to diagnose ACS. **[2010]**

1.2.6.8 Consider a chest X-ray to help exclude complications of ACS such as pulmonary oedema, or other diagnoses such as pneumothorax or pneumonia. **[2010]**

- 1.2.6.9 If an ACS has been excluded at any point in the care pathway, but people have risk factors for cardiovascular disease, follow the appropriate guidance, for example, the [NICE guideline on cardiovascular disease](#) and the [NICE guideline on hypertension in adults](#). **[2010]**

1.3 People presenting with stable chest pain

This section of the guideline addresses the assessment and diagnosis of intermittent stable chest pain in people with suspected stable angina.

- 1.3.1.1 Exclude a diagnosis of stable angina if clinical assessment indicates non-anginal chest pain (see recommendation 1.3.3.1) and there are no other aspects of the history or risk factors raising clinical suspicion. **[2016]**
- 1.3.1.2 If clinical assessment indicates typical or atypical angina (see recommendation 1.3.3.1), offer diagnostic testing (see the [sections on diagnostic testing for people in whom stable angina cannot be excluded by clinical assessment alone, additional diagnostic investigations and use of non-invasive functional testing for myocardial ischaemia](#)). **[2016]**

1.3.2 Clinical assessment

- 1.3.2.1 Take a detailed clinical history documenting:
- the age and sex of the person
 - the characteristics of the pain, including its location, radiation, severity, duration and frequency, and factors that provoke and relieve the pain
 - any associated symptoms, such as breathlessness
 - any history of angina, MI, coronary revascularisation or other cardiovascular disease **and**
 - any cardiovascular risk factors. **[2010]**

1.3.2.2 Carry out a physical examination to:

- identify risk factors for cardiovascular disease
- identify signs of other cardiovascular disease
- identify non-coronary causes of angina (for example, severe aortic stenosis, cardiomyopathy) **and**
- exclude other causes of chest pain. **[2010]**

1.3.3 Making a diagnosis based on clinical assessment

1.3.3.1 Assess the typicality of chest pain as follows:

- Presence of three of the features below is defined as typical angina.
- Presence of two of the three features below is defined as atypical angina.
- Presence of one or none of the features below is defined as non-anginal chest pain.

Anginal pain is:

- constricting discomfort in the front of the chest, or in the neck, shoulders, jaw or arms
- precipitated by physical exertion
- relieved by rest or GTN within about 5 minutes. **[2010, amended 2016]**

1.3.3.2 Do not define typical and atypical features of anginal chest pain and non-anginal chest pain differently in men and women. **[2010]**

1.3.3.3 Do not define typical and atypical features of anginal chest pain and non-anginal chest pain differently in ethnic groups. **[2010]**

1.3.3.4 Take the following factors, which make a diagnosis of stable angina more likely, into account when estimating people's likelihood of angina:

- age
- whether the person is male
- cardiovascular risk factors including:
 - a history of smoking
 - diabetes
 - hypertension
 - dyslipidaemia
 - family history of premature coronary artery disease (CAD)

other cardiovascular disease

- history of established CAD, for example, previous MI, coronary revascularisation. **[2010]**

1.3.3.5 Unless clinical suspicion is raised based on other aspects of the history and risk factors, exclude a diagnosis of stable angina if the pain is non-anginal (see recommendation 1.3.3.1). Features which make a diagnosis of stable angina unlikely are when the chest pain is:

- continuous or very prolonged **and/or**
- unrelated to activity **and/or**
- brought on by breathing in **and/or**
- associated with symptoms such as dizziness, palpitations, tingling or difficulty swallowing.

Consider causes of chest pain other than angina (such as gastrointestinal or musculoskeletal pain). **[2010]**

1.3.3.6 Consider investigating other causes of angina, such as hypertrophic cardiomyopathy, in people with typical angina-like chest pain and a low likelihood of CAD. **[2010, amended 2016]**

- 1.3.3.7 Arrange blood tests to identify conditions which exacerbate angina, such as anaemia, for all people being investigated for stable angina. **[2010]**
- 1.3.3.8 Only consider chest X-ray if other diagnoses, such as a lung tumour, are suspected. **[2010]**
- 1.3.3.9 If a diagnosis of stable angina has been excluded at any point in the care pathway, but people have risk factors for cardiovascular disease, follow the appropriate guidance, for example, the [NICE guideline on cardiovascular disease](#) and the [NICE guideline on hypertension in adults](#). **[2010]**
- 1.3.3.10 For people in whom stable angina cannot be excluded on the basis of the clinical assessment alone, take a resting 12-lead ECG as soon as possible after presentation. **[2010, amended 2016]**
- 1.3.3.11 Do not rule out a diagnosis of stable angina on the basis of a normal resting 12-lead ECG. **[2010]**
- 1.3.3.12 Do not offer diagnostic testing to people with non-anginal chest pain on clinical assessment (see recommendation 1.3.3.1) unless there are resting ECG ST-T changes or Q waves. **[2016]**
- 1.3.3.13 A number of changes on a resting 12-lead ECG are consistent with CAD and may indicate ischaemia or previous infarction. These include:
- pathological Q waves in particular
 - LBBB
 - ST-segment and T wave abnormalities (for example, flattening or inversion).
- Note that the results may not be conclusive.
- Consider any resting 12-lead ECG changes together with people's clinical history and risk factors. **[2010]**
- 1.3.3.14 For people with confirmed CAD (for example, previous MI, revascularisation, previous angiography) in whom stable angina cannot be excluded based on

clinical assessment alone, see recommendation 1.3.4.4 about functional testing. **[2010, amended 2016]**

- 1.3.3.15 Consider aspirin only if the person's chest pain is likely to be stable angina, until a diagnosis is made. Do not offer additional aspirin if there is clear evidence that people are already taking aspirin regularly or are allergic to it. **[2010]**
- 1.3.3.16 Follow the [NICE guideline on stable angina](#) while waiting for the results of investigations if symptoms are typical of stable angina. **[2010]**

1.3.4 Diagnostic testing for people in whom stable angina cannot be excluded by clinical assessment alone

The Guideline Development Group emphasised that the recommendations in this guideline are to make a diagnosis of chest pain, not to screen for CAD. Most people diagnosed with non-anginal chest pain after clinical assessment need no further diagnostic testing. However in a very small number of people, there are remaining concerns that the pain could be ischaemic.

- 1.3.4.1 Include the typicality of anginal pain features (see recommendation 1.3.3.1) in all requests for diagnostic investigations and in the person's notes. **[2010, amended 2016]**
- 1.3.4.2 Use clinical judgement and take into account people's preferences and comorbidities when considering diagnostic testing. **[2010]**
- 1.3.4.3 Offer 64-slice (or above) CT coronary angiography if:
- clinical assessment (see recommendation 1.3.3.1) indicates typical or atypical angina **or**
 - clinical assessment indicates non-anginal chest pain but 12-lead resting ECG has been done and indicates ST-T changes or Q waves. **[2016]**
- 1.3.4.4 For people with confirmed CAD (for example, previous MI, revascularisation, previous angiography), offer non-invasive functional testing when there is uncertainty about whether chest pain is caused by myocardial ischaemia. See the

[section on use of non-invasive functional testing for myocardial ischaemia](#) for further guidance on non-invasive functional testing. An exercise ECG may be used instead of functional imaging. **[2010]**

See also [NICE's medical technologies guidance on HeartFlow FFRCT for estimating fractional flow reserve from coronary CT angiography](#). Using HeartFlow FFR_{CT} may avoid the need for invasive coronary angiography and revascularisation for some patients

1.3.5 Additional diagnostic investigations

- 1.3.5.1 Offer non-invasive functional imaging (see the [section on use of non-invasive functional testing for myocardial ischaemia](#)) for myocardial ischaemia if 64-slice (or above) CT coronary angiography has shown CAD of uncertain functional significance or is non-diagnostic. **[2016]**
- 1.3.5.2 Offer invasive coronary angiography as a third-line investigation when the results of non-invasive functional imaging are inconclusive. **[2016]**

1.3.6 Use of non-invasive functional testing for myocardial ischaemia

- 1.3.6.1 When offering non-invasive functional imaging for myocardial ischaemia use:
- myocardial perfusion scintigraphy with single photon emission computed tomography (MPS with SPECT) **or**
 - stress echocardiography **or**
 - first-pass contrast-enhanced magnetic resonance (MR) perfusion **or**
 - MR imaging for stress-induced wall motion abnormalities.

Take account of locally available technology and expertise, the person and their preferences, and any contraindications (for example, disabilities, frailty, limited ability to exercise) when deciding on the imaging method. **[This**

recommendation updates and replaces recommendation 1.1 of NICE's technology appraisal guidance on myocardial perfusion scintigraphy for the diagnosis and management of angina and myocardial infarction]. **[2016]**

- 1.3.6.2 Use adenosine, dipyridamole or dobutamine as stress agents for MPS with SPECT and adenosine or dipyridamole for first-pass contrast-enhanced MR perfusion. **[2010]**
- 1.3.6.3 Use exercise or dobutamine for stress echocardiography or MR imaging for stress-induced wall motion abnormalities. **[2010]**
- 1.3.6.4 Do not use MR coronary angiography for diagnosing stable angina. **[2010]**
- 1.3.6.5 Do not use exercise ECG to diagnose or exclude stable angina for people without known CAD. **[2010]**

1.3.7 Making a diagnosis following investigations

Box 1 Definition of significant coronary artery disease

Significant coronary artery disease (CAD) found during CT coronary angiography is $\geq 70\%$ diameter stenosis of at least one major epicardial artery segment or $\geq 50\%$ diameter stenosis in the left main coronary artery:

Factors intensifying ischaemia

Such factors allow less severe lesions (for example, $\geq 50\%$) to produce angina:

- reduced oxygen delivery: anaemia, coronary spasm
- increased oxygen demand: tachycardia, left ventricular hypertrophy
- large mass of ischaemic myocardium: proximally located lesions
- longer lesion length.

Factors reducing ischaemia which may render severe lesions ($\geq 70\%$) asymptomatic:

- Well-developed collateral supply.
- Small mass of ischaemic myocardium: distally located lesions, old infarction in the territory of coronary supply. **[2016]**

1.3.7.1 Confirm a diagnosis of stable angina and follow the [NICE guideline on stable angina](#) when:

- significant CAD (see box 1) is found during invasive or 64-slice (or above) CT coronary angiography **or**
- reversible myocardial ischaemia is found during non-invasive functional imaging. **[2016]**

1.3.7.2 Investigate other causes of chest pain when:

- significant CAD (see box 1) is not found during invasive coronary angiography or 64-slice (or above) CT coronary angiography **or**
- reversible myocardial ischaemia is not found during non-invasive functional imaging. **[2016]**

1.3.7.3 Consider investigating other causes of angina, such as hypertrophic cardiomyopathy or syndrome X, in people with typical angina-like chest pain if investigation excludes flow-limiting disease in the epicardial coronary arteries. **[2010]**

Terms used in this guideline

Chest pain

The term 'chest pain' is used throughout the guideline to mean chest pain or discomfort.

Recommendations for research

In 2010, the guideline committee made the following recommendations for research. The committee's full set of recommendations for research is detailed in the [full guideline](#).

1 Cost-effectiveness of multislice CT coronary angiography for ruling out obstructive coronary artery disease in people with troponin-negative acute coronary syndromes

Is multislice CT coronary angiography a cost-effective first-line test for ruling out obstructive coronary artery disease (CAD) in people with suspected troponin-negative acute coronary syndromes?

Why the committee made the recommendation for research

Current European Society of Cardiology guidelines state that in troponin-negative ACS with no ST-segment change on the ECG, 'a stress test is recommended... in patients with significant ischaemia during the stress test, coronary angiography and subsequent revascularisation should be considered'. Yet stress testing has relatively low sensitivity and specificity for diagnosing CAD in this group of people. Therefore a significant proportion of at-risk people are missed while others with normal coronary arteries are subjected to an unnecessary invasive coronary angiogram. Multislice CT coronary angiography is highly sensitive and provides a potentially useful means for early rule-out of CAD in troponin-negative acute coronary disease. We need to know whether it is cost effective compared with exercise ECG as a first test in the diagnostic work-up of this group.

Full details of the evidence and committee's discussion are in the [full guideline](#).

2 Refining the use of telephone advice in people with chest pain

In what circumstances should telephone advice be given to people calling with chest pain? Is the appropriateness influenced by age, sex or symptoms?

Why the committee made the recommendation for research

The telephone is a common method of first contact with healthcare services, and produces a near uniform emergency response to chest pain symptoms. Such a response has considerable economic, social and human costs. Research should be conducted to clarify if an emergency response in all circumstances is appropriate, or if there are identifiable factors such as age, sex or associated symptoms that would allow a modified response and a more appropriate use of resources.

Full details of the evidence and committee's discussion are in the [full guideline](#).

3 Establishing a national registry for people who are undergoing initial assessment for stable angina

Can a national registry of people presenting with suspected angina be established to allow cohort analysis of treatments, investigations and outcomes in this group? Such a registry would provide a vital resource for a range of important research projects, including:

- development and validation of a new score for assessing the pre-test probability of disease, addressing outstanding uncertainties in the estimation of the pre-test probability of CAD based on simple measures made at initial assessment (history, examination, routine bloods, resting 12-lead ECG)
- assessment of the extent to which new circulating biomarkers add additional information to measures made at initial assessment
- provision of a framework for trial recruitment without significant work-up bias allowing evaluation of the diagnostic and prognostic test performance of CT-based, MR, echocardiography and radionuclide technologies.

Why the committee made the recommendation for research

A national prospective registry of consecutive people with suspected stable angina before initial diagnostic testing does not currently exist in the UK or in any other country. Establishing such a registry would offer the following methodological strengths: statistical size, representative patients without work-up bias, contemporary data. This would overcome key problems in much of the existing evidence base.

Accurate assessment of pre-test likelihood of coronary disease is needed to inform the cost-effective choice of investigative technologies such as CT coronary calcium scoring for people with chest pain that may be caused by myocardial ischaemia. The data on which pre-test likelihood is based date from 1979 in a US population and may not be applicable to contemporary UK populations. There remain continuing uncertainties about the initial assessment of people with suspected stable angina. For example, the possible contributions of simple clinical measures such as body mass index, routine blood markers (for example, haemoglobin) or novel circulating biomarkers to estimates of the pre-test likelihood of CAD are not known and require further assessment in the whole population and in predefined subgroups including ethnic minorities.

Full details of the evidence and committee's discussion are in the [full guideline](#).

4 Information about presenting and explaining tests

All people presenting with chest pain will need to decide whether to accept the diagnostic and care pathways offered. How should information about the diagnostic pathway and the likely outcomes, risks and benefits, with and without treatment, be most effectively presented to particular groups of people, defined by age, ethnicity and sex?

Why the committee made the recommendation for research

Methods of communication (both the content and delivery) will be guided by current evidence-based best practice. Controlled trials should be conducted based on well-constructed randomised controlled clinical trials comparing the effects of different methods of communication on the understanding of the person with chest pain. Such studies might consider a number of delivery mechanisms, including advice and discussion with a clinician or a specialist nurse, as well as specific information leaflets or visual data.

Any trials should also investigate the feasibility of introducing a suggested guideline protocol to be used with all people presenting with chest pain when faced with options concerning their clinical pathway.

Only by clearly explaining and then discussing the proposed diagnostic and care pathways can the healthcare professional be reasonably certain that informed consent has been obtained and that a patient's moral, ethical and spiritual beliefs, expectations, and any misconceptions about their condition, have been taken into account. Consideration should be given to any communication problems the person may have.

Full details of the evidence and committee's discussion are in the [full guideline](#).

Putting this guideline into practice

NICE has produced [tools and resources](#) to help you put this guideline into practice.

Putting recommendations into practice can take time. How long may vary from guideline to guideline, and depends on how much change in practice or services is needed. Implementing change is most effective when aligned with local priorities.

Changes recommended for clinical practice that can be done quickly – like changes in prescribing practice – should be shared quickly. This is because healthcare professionals should use guidelines to guide their work – as is required by professional regulating bodies such as the General Medical and Nursing and Midwifery Councils.

Changes should be implemented as soon as possible, unless there is a good reason for not doing so (for example, if it would be better value for money if a package of recommendations were all implemented at once).

Different organisations may need different approaches to implementation, depending on their size and function. Sometimes individual practitioners may be able to respond to recommendations to improve their practice more quickly than large organisations.

Here are some pointers to help organisations put NICE guidelines into practice:

1. **Raise awareness** through routine communication channels, such as email or newsletters, regular meetings, internal staff briefings and other communications with all relevant partner organisations. Identify things staff can include in their own practice straight away.
2. **Identify a lead** with an interest in the topic to champion the guideline and motivate others to support its use and make service changes, and to find out any significant issues locally.
3. **Carry out a baseline assessment** against the recommendations to find out whether there are gaps in current service provision.
4. **Think about what data you need to measure improvement** and plan how you will collect it. You may want to work with other health and social care organisations and specialist

groups to compare current practice with the recommendations. This may also help identify local issues that will slow or prevent implementation.

5. Develop an action plan, with the steps needed to put the guideline into practice, and make sure it is ready as soon as possible. Big, complex changes may take longer to implement, but some may be quick and easy to do. An action plan will help in both cases.

6. For very big changes include milestones and a business case, which will set out additional costs, savings and possible areas for disinvestment. A small project group could develop the action plan. The group might include the guideline champion, a senior organisational sponsor, staff involved in the associated services, finance and information professionals.

7. Implement the action plan with oversight from the lead and the project group. Big projects may also need project management support.

8. Review and monitor how well the guideline is being implemented through the project group. Share progress with those involved in making improvements, as well as relevant boards and local partners.

NICE provides a comprehensive programme of support and resources to maximise uptake and use of evidence and guidance. See our [into practice pages](#) for more information.

Also see [Leng G, Moore V, Abraham S, editors \(2014\) Achieving high quality care – practical experience from NICE](#). Chichester: Wiley.

Context

Conditions causing chest pain or discomfort, such as an acute coronary syndrome or angina, have a potentially poor prognosis, emphasising the importance of prompt and accurate diagnosis. Treatments are available to improve symptoms and prolong life, hence the need for this guideline.

This guideline covers the assessment and diagnosis of recent onset chest pain or discomfort of suspected cardiac origin. In deciding whether chest pain may be cardiac and therefore whether this guideline is relevant, a number of factors should be taken into account. These include the person's history of chest pain, their cardiovascular risk factors, history of ischaemic heart disease and any previous treatment, and previous investigations for chest pain.

For pain that is suspected to be cardiac, there are two separate diagnostic pathways presented in the guideline. The first is for people with acute chest pain and a suspected acute coronary syndrome, and the second is for people with intermittent stable chest pain in whom stable angina is suspected. The guideline includes how to determine whether myocardial ischaemia is the cause of the chest pain and how to manage the chest pain while people are being assessed and investigated.

As far as possible, the recommendations in this guideline have been listed in the order in which they will be carried out and follow the diagnostic pathways. But, as there are many permutations at each decision point, it has been necessary to include frequent cross-referencing to avoid repeating recommendations several times.

The algorithms presented in [full guideline](#) show the two diagnostic pathways.

This guideline does not cover the diagnosis and management of chest pain that is unrelated to the heart (for example, traumatic chest wall injury, herpes zoster infection) when myocardial ischaemia has been excluded. The guideline also recognises that in people with a prior diagnosis of coronary artery disease, chest pain or discomfort is not necessarily cardiac.

The term 'chest pain' is used throughout the guideline to mean chest pain or discomfort.

The guideline will assume that prescribers will use a drug's summary of product

characteristics to inform decisions made with individual patients.

Finding more information and committee details

To find NICE guidance on related topics, including guidance in development, see the [NICE topic page on cardiovascular conditions](#).

For full details of the evidence and the guideline committee's discussions, see the [full guideline](#). You can also find information about [how this guideline was developed](#), including details of the committee.

NICE has produced [tools and resources to help you put this guideline into practice](#). For general help and advice on putting our guidelines into practice, see [resources to help you put NICE guidance into practice](#).

Update information

November 2016: New recommendations have been added for the assessment and diagnosis of acute chest pain and the assessment and diagnosis of stable chest pain.

These are marked as **[2016]**.

Where recommendations end **[2010]**, the evidence has not been reviewed since the original guideline.

We have also made some changes without an evidence review:

- some 2010 recommendations have been updated to align with the 2016 recommendations
- some recommendations have been updated to clarify the use of high-sensitivity troponin testing
- some recommendations have been updated to remove reference to estimate of the likelihood of coronary artery disease
- reference to autopsy as a diagnostic criterion has been removed.

These recommendations are marked **[2010, amended 2016]**.

Minor changes since publication

October 2022: We added text to indicate that pulse oximetry may be less reliable in people with dark skin. We also added a link to the NHS patient safety alert on the risk of harm from inappropriate placement of pulse oximeter probes. See recommendation 1.2.3.3.

November 2020: We linked to the updated guideline on acute coronary syndromes.

August 2020: We linked to updated diagnostics guidance in recommendations on use of biochemical markers for diagnosis (section 1.2.5).

January 2020: The title of the guideline was updated to clarify that it covers chest pain of suspected cardiac origin. We linked to the universal definition of myocardial infarction in

recommendation 1.2.6.1 and added a cross-reference to related NICE medical technologies guidance at the end of section 1.3.4.

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